

Biological Age Estimation from Chest Radiographs Using Deep Learning: A ResNet-50 Based Approach with Clinical Risk Stratification

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Abstract—Biological age frequently provides a more accurate representation of an individual’s physiological condition than chronological age and can serve as a valuable indicator for identifying individuals at an elevated risk of age-associated diseases. Yet many established biological-age pipelines depend on multiple clinical inputs, including blood biomarkers, genomic assays, and detailed medical records, which makes large-scale deployment difficult, especially in resource-limited settings. This paper introduces BioAge-Net, a deep learning framework that predicts biological age directly from chest radiographs using a ResNet-50 backbone and a custom regression head trained end-to-end on the NIH ChestX-Ray14 dataset (112,104 images from 30,805 patients). Each radiograph is resized to 224×224 and preprocessed with Gaussian blur, CLAHE, and ImageNet normalization. Global-average pooled ResNet-50 features are then passed through fully connected layers ($2048 \rightarrow 512 \rightarrow 128 \rightarrow 1$) with batch normalization and dropout, and the network is optimized with mean squared error loss. On 25,592 held-out test images, the model achieves an MAE of 4.91 years, RMSE of 6.43 years, R^2 of 0.8455, and Pearson r of 0.9198 ($p < 0.001$), with a mean bias of +0.24 years. Using the biological age gap (predicted age minus chronological age), patients are grouped into Low (18.51%), Moderate (60.55%), and High (20.94%) risk categories. Grad-CAM heatmaps provide visual evidence of the regions driving the predictions. Overall, the findings suggest that a single chest X-ray can support practical, non-invasive biological age estimation in scalable screening workflows.

Index Terms—Biological age estimation, chest radiograph, deep learning, ResNet-50, Grad-CAM, risk stratification, transfer learning, NIH ChestX-Ray14

I. INTRODUCTION

Aging is not a uniform process: two people of the same chronological age can differ markedly in physiological reserve, disease burden, and future health risk. Biological age is intended to capture this functional state more directly than date of birth and often aligns more closely with clinical outcomes than chronological age alone [1], [2]. When biological age exceeds chronological age, the risks of mortality and other adverse aging-related outcomes generally increase [3].

Most current approaches to biological-age estimation depend on multimodal clinical data, including blood biochemistry, DNA methylation measures such as Horvath’s epigenetic

clock, telomere-length assessment, and phenotypic scoring systems [4], [5]. Although these approaches are well established, they require time, infrastructure, and technical expertise, which can limit routine screening in resource-constrained settings.

Chest radiographs are among the most commonly used diagnostic X-ray studies in medicine [6]. A standard posteroanterior (PA) chest X-ray captures a broad range of anatomical cues, including cardiothoracic contours, bone density, soft-tissue distribution, and vascular patterns, that change with age. Even so, chest radiographs are still underused as a direct source of biological-age information.

At the same time, deep convolutional neural networks (CNNs) and large annotated imaging datasets have made it increasingly feasible to learn radiographic biomarkers automatically. Transfer learning with ImageNet-pretrained backbones such as ResNet has produced strong results in disease-classification tasks and has motivated broader adoption across medical imaging applications [7], [8].

This work presents BioAge-Net, an end-to-end framework for estimating biological age directly from chest radiographs.

- 1) An end-to-end deep learning pipeline is built for biological-age regression from chest radiographs and trained on the largest publicly available chest X-ray dataset used in this study (NIH ChestX-Ray14, $n = 112,104$).
- 2) Competitive single-modality performance is obtained, with MAE = 4.91 years, $R^2 = 0.8455$, and Pearson $r = 0.9198$, without using auxiliary clinical variables.
- 3) A clinically oriented three-level risk stratification scheme is defined from the biological age gap so that model outputs can be interpreted more easily in screening workflows.
- 4) Grad-CAM visualizations are included to make the learned radiographic focus regions more interpretable for clinical AI deployment.

The rest of the paper is organized as follows. Section II reviews the related literature. Section III describes the dataset and preprocessing pipeline. Section IV presents the proposed

architecture. Section V reports the experimental results. Section VI concludes the paper and outlines future directions.

II. RELATED WORK

A. Classical Biological-Age Methods

Levine et al. [3] created a way to measure phenotypic age using nine blood-related clinical markers. They showed that the difference between a person’s biological age and actual age can predict the risk of dying from any cause. Horvath’s DNA methylation clock [4] calculates biological age by looking at how DNA is methylated. This method has an average error of about 3.6 years when applied to blood samples, making it one of the most commonly referenced standards in the field. Klemra and Doubal [5] introduced a composite regression model designed to minimize the discrepancy between chronological age and biomarker-derived age estimates. While these methodologies are both influential and often exhibit high accuracy, their reliance on non-imaging biomarkers limits their applicability in screening protocols that primarily utilize routine radiographic imaging.

B. Deep Learning for Medical Age Estimation

Age estimation with deep learning has also been studied across multiple imaging modalities. Iglovikov et al. [9] applied CNNs to pediatric hand and wrist radiographs, while Rim et al. [10] used retinal fundus photographs to predict systemic biomarkers, including age and sex, with deep-learning models. Raghu et al. [11] examined transfer-learning behavior in medical imaging and motivated careful backbone selection, and later Raghu et al. [12] showed that age estimates derived from chest radiographs carry meaningful cardiovascular prognostic value. Ieki et al. [13] further demonstrated that the chest X-ray age gap independently predicts cardiovascular outcomes, reinforcing the clinical relevance of radiographic aging signals, while architectures such as DenseNet-121, EfficientNet, and Vision Transformers have continued to show strong performance across related chest X-ray tasks [7], [14], [15].

C. NIH ChestX-Ray14 and Research Gap

The NIH ChestX-Ray14 dataset used in this study extends the ChestX-ray8 benchmark introduced by Wang et al. [16] and, according to the NIH dataset release [17], includes 112,120 frontal-view chest radiographs with demographic metadata such as age and sex. Despite its widespread use in disease classification, comparatively few studies have used it for biological-age regression. Despite progress in multimodal aging biomarkers and chest X-ray deep learning, the literature still lacks an integrated end-to-end pipeline that combines (a) biological age regression, (b) clinically interpretable risk grouping, and (c) explainable AI using chest radiographs alone at the scale of more than 100,000 images.

III. DATASET AND PREPROCESSING

A. Dataset

All the experiments in this study were done using the NIH ChestX-Ray14 dataset [17]. This dataset includes 112,120 front-view chest X-rays from 30,805 different patients who

were scanned at the NIH Clinical Center between 1992 and 2015. After removing 16 records that had ages that did not make sense based on the patients’ medical information, 112,104 valid samples were kept.

TABLE I
DATASET STATISTICS

Property	Value
Total samples	112,104
Unique patients	30,805
Age range	1–95 years
Mean age $\pm$ std	46.87 ± 16.60 years
Male / Female	63,328 (56.5%) / 48,776 (43.5%)
Training samples	77,860 (69.4%)
Validation samples	8,652 (7.7%)
Test samples	25,592 (22.8%)

The official split from NIH into training and testing sets, as shown in Table I, was followed. Then, the training set was further divided into training and validation parts using a 90/10 ratio. The age distribution stayed similar across all three groups: Train (average age 46.8 years), Val (average age 46.9 years), and Test (average age 47.0 years).

B. Preprocessing Pipeline

Before both training and testing, every chest X-ray goes through a standard set of steps to prepare it. These steps are:

- 1) Gaussian Blur: A 3×3 Gaussian filter with a standard deviation of 1.0 is used to reduce high-frequency noise from the image. This helps make the later steps of feature detection more accurate and stable.
- 2) CLAHE (Contrast Limited Adaptive Histogram Equalization) [18]: This is done with a clip limit of 2.0 and a tile grid size of 8×8 . It improves the contrast in different parts of the image, especially in areas where there is not much variation in brightness. This helps in seeing subtle differences such as changes in bone density or calcifications in blood vessels.
- 3) Resizing: All images are made the same size, 224×224 pixels, to match what the ResNet-50 model expects as input.
- 4) Channel Stacking: Each image, which is in grayscale, is copied three times to create a 3-channel image. This makes it compatible with the pre-trained ImageNet model.
- 5) Normalization: Each channel is adjusted using the mean and standard deviation values from the ImageNet dataset. The mean values are [0.485, 0.456, 0.406] and the standard deviations are [0.229, 0.224, 0.225].
- 6) Training Augmentation: During training, extra steps are added to make the model more flexible. These include flipping the image horizontally with a 50% chance, rotating it by up to 10 degrees, and slightly changing the brightness.

After these steps, the images are converted into float32 tensors with dimensions (B, 3, 224, 224), where B is the number of images in a batch.

IV. PROPOSED ARCHITECTURE

A. Overview

The BioAge-Net architecture is summarized in Fig. 1. It consists of four connected components: a ResNet-50 feature extractor, a custom regression head, a clinical risk stratification module, and a Grad-CAM explainability layer.

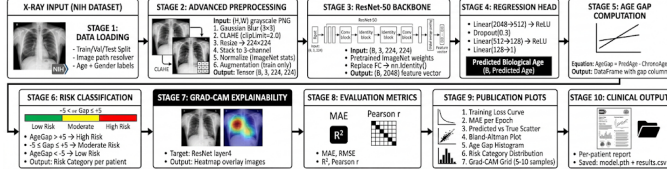


Fig. 1. BioAge-Net architecture overview.

B. Feature Extractor: ResNet-50

ResNet-50 [8], pretrained on ImageNet [19] (1.28M images, 1000 classes), serves as the feature-extraction backbone. Through residual (skip) connections and 16 bottleneck blocks arranged in four stages, it learns progressively more abstract spatial representations from the input radiograph.

The original 1000-class classification head is removed and replaced with `nn.Identity()`, which outputs a 2048-dimensional global-average-pooled feature vector. All 23,508,032 backbone parameters are fine-tuned so that the pretrained natural-image features adapt to age-related radiographic patterns.

Input: $(B, 3, 224, 224)$ normalized chest radiograph tensor

Output: $(B, 2048)$ feature embedding

C. Regression Head

The regression head converts the 2048-dimensional embedding into a scalar biological age estimate through three regularized fully connected layers, as shown in Fig. 2:

The full model has 24,622,913 trainable parameters (Table II). Batch normalization helps stabilize optimization during fine-tuning, while dropout reduces overfitting by limiting feature co-adaptation.

D. Training Configuration

This subsection summarizes the optimization strategy used to train BioAge-Net end to end. The selected hyperparameters were chosen to promote stable convergence, reduce overfitting

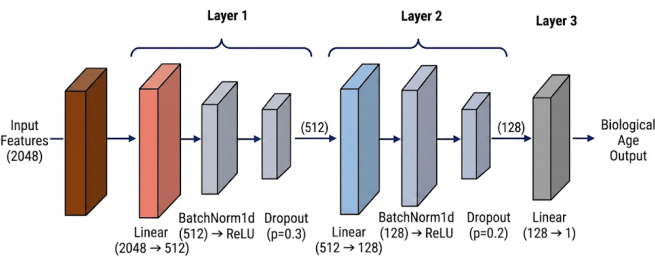


Fig. 2. Regression head architecture.

TABLE II
ARCHITECTURE SUMMARY.

Component	Output Shape	Parameters
Input	$B \times 3 \times 224 \times 224$	—
ResNet-50 (backbone)	$B \times 2048$	23,508,032
FC(2048→512) + BN + ReLU +	$B \times 512$	1,049,600
Drop		
FC(512→128) + BN + ReLU +	$B \times 128$	65,792
Drop		
FC(128→1)	$B \times 1$	129
Total	—	24,622,913

during fine-tuning, and preserve generalization on unseen chest radiographs. Table III lists the main training hyperparameters.

TABLE III
TRAINING HYPERPARAMETERS.

Hyperparameter	Value
Loss function	Mean squared error (MSE)
Optimizer	Adam [20] ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial LR	1×10^{-4}
LR schedule	ReduceLROnPlateau (factor = 0.5, patience = 3)
Minimum LR	2.5×10^{-5} (reached at epoch 19)
Batch size	32
Epochs	20
Hardware	2 × Tesla T4 GPUs (DataParallel)
Total training time	~435 min

E. Risk Stratification Module

After the model predicts biological age, an age-gap score is computed:

$$\Delta \text{Age} = \text{Predicted Biological Age} - \text{Chronological Age} \quad (1)$$

Using the age-gap definition in (1), each patient is assigned to one of three clinical risk groups:

Low risk: $\Delta \text{Age} < -5$ years

Biological age is more than 5 years lower than chronological age, suggesting relatively preserved physiological health.

Moderate risk: $-5 \leq \Delta \text{Age} \leq +5$ years

Biological and chronological ages are broadly aligned, so routine preventive screening remains appropriate.

High risk: $\Delta \text{Age} > +5$ years

Biological age is more than 5 years higher than chronological age, suggesting accelerated aging and the need for closer clinical monitoring.

F. Grad-CAM Explainability

To interpret model behavior, Gradient-weighted Class Activation Mapping (Grad-CAM) [21] was applied to ResNet-50's final convolutional block (layer4) to generate spatial saliency maps. The gradients of the predicted age output with respect to the layer4 activations are global-average-pooled to compute channel-importance weights, which are then combined with the feature maps and passed through ReLU to form the heatmap.

The resulting saliency maps are bilinearly upsampled to 224×224 and overlaid on the original radiographs with a

jet colormap. In practice, the maps highlight the regions that contribute most strongly to age estimation, including the thoracic spine, cardiac silhouette, and rib cortex. Representative Grad-CAM examples are shown in Fig. 3.

V. EXPERIMENTAL RESULTS

A. Quantitative Performance

Quantitative results are summarized in Table IV.

TABLE IV
HELD-OUT TEST-SET PERFORMANCE AND BEST VALIDATION-CHECKPOINT METRICS.

Metric	Value
MAE	4.9108 years
RMSE	6.4340 years
R^2 score	0.8455
Pearson r	0.9198 ($p < 0.001$)
Mean bias	+0.2416 years
Std bias	6.4295 years
Best val MAE	3.6170 years (Epoch 20)
Best val R^2	0.9180 (Epoch 20)

On the held-out test set of 25,592 radiographs, BioAge-Net achieved an MAE of 4.91 years. The combination of a strong Pearson correlation ($r = 0.9198$, $p < 0.001$) and $R^2 = 0.8455$ suggests that the model captures much of the age-related signal present in the radiographs without relying on auxiliary clinical data. The minimal mean bias of +0.24 years further demonstrates robust calibration at the population level, indicating only a slight tendency toward systematic overestimation or underestimation.

B. Training Convergence

The training process stayed stable over 20 epochs, as shown in Fig. 4. The learning rate was lowered at epochs 12 and 19, and both times there was an improvement in how well the model performed on validation data. The difference between the training and validation MAE stayed below one year during the whole training process, suggesting that the model did not overfit much.

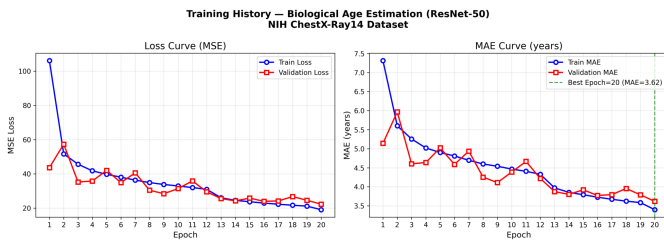


Fig. 4. Training and validation curves for BioAge-Net.

C. Biological Age Gap Analysis

As shown in Fig. 5, the age differences follow a normal distribution, centered close to zero, with an average of +0.24 years. This result matches what we would expect from accurate population data. Most of the data points are near zero, while only a small number show large positive or negative differences.

TABLE V
AGE-GAP STATISTICS ON THE TEST SET.

Statistic	Value
Mean gap	+0.2416 yrs
Std gap	6.4296 yrs
Median gap	+0.3554 yrs
Min gap	-50.34 yrs
Max gap	+60.34 yrs

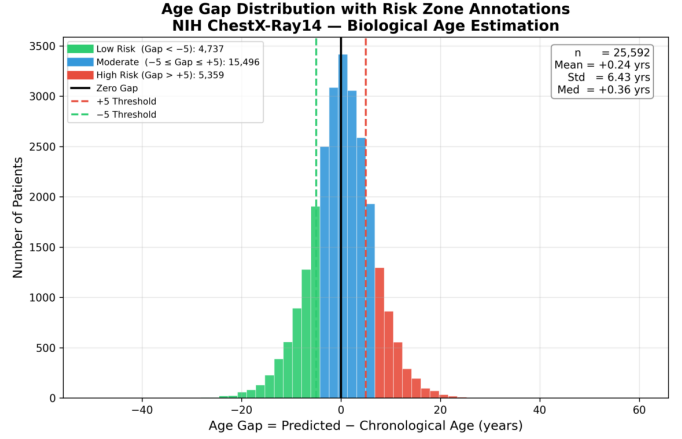


Fig. 5. Biological age gap distribution on the test set.

D. Risk Stratification Results

TABLE VI
RISK CATEGORY DISTRIBUTION ON THE TEST SET.

Category	Count	Percentage
Low risk	4,737	18.51%
Moderate risk	15,496	60.55%
High risk	5,359	20.94%
Total	25,592	100.00%

Most of the patients, about 60.55%, were put into the moderate-risk group. This shows that their predicted biological age is nearly the same as their actual age. Another 18.51% were in the low-risk group, meaning their biological age is younger than their real age, which suggests they are in better health. The rest, around 20.94%, were in the high-risk group, showing signs of faster aging, which might mean they need more careful check-ups in this hospital cohort with many different illnesses (see Table VI).

E. Cross-Dataset Validation on CheXpert

To assess external generalizability, BioAge-Net was evaluated on the CheXpert validation set [22], which comprises 202 frontal chest radiographs from a held-out Stanford institutional cohort, without any fine-tuning or retraining. The same preprocessing pipeline (Gaussian blur, CLAHE, ImageNet normalization) was applied, and the results are summarized in Table VII.

The drop in performance is consistent with the expected domain shift across institutions, imaging equipment, and patient demographics (CheXpert mean age: 60.8 years vs. NIH

Grad-CAM Explainability — Biological Age Estimation (ResNet-50)
Row 1: Original X-Ray | Row 2: Grad-CAM Heatmap Overlay

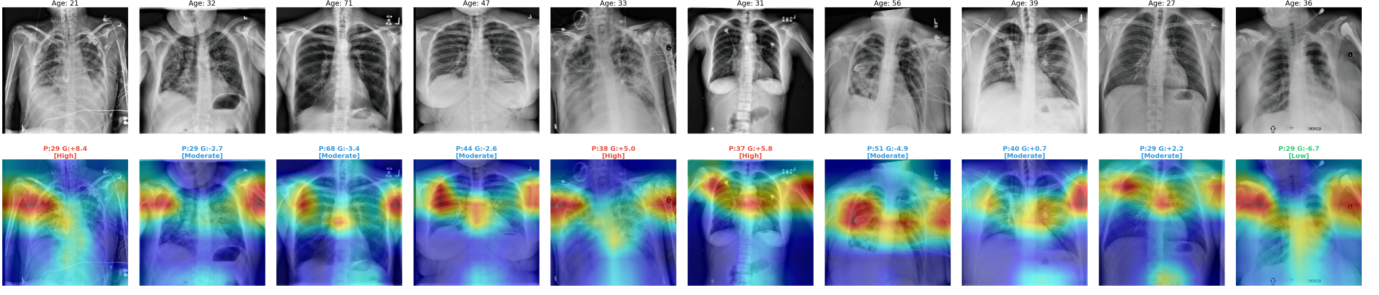


Fig. 3. Grad-CAM saliency maps from ResNet-50’s final convolutional block for representative test cases across the three risk categories. P = predicted biological age; G = biological age gap. The strongest activations appear in clinically meaningful regions, including the cardiac silhouette, thoracic spine, and rib cortex, which are consistent with known radiographic aging biomarkers.

TABLE VII
CROSS-DATASET VALIDATION RESULTS ON CHEXPert.

Metric	NIH (in-domain)	CheXpert (cross-domain)
MAE	4.91 years	7.28 years
RMSE	6.43 years	9.03 years
R^2	0.8455	0.7564
Pearson r	0.9198	0.8834
Mean Bias	+0.24 yrs	+0.21 yrs

mean age: 46.9 years). Even under zero-shot transfer, the radiographic aging signal learned from NIH data remains meaningful on an independent institutional dataset, as further illustrated in Fig. 6.

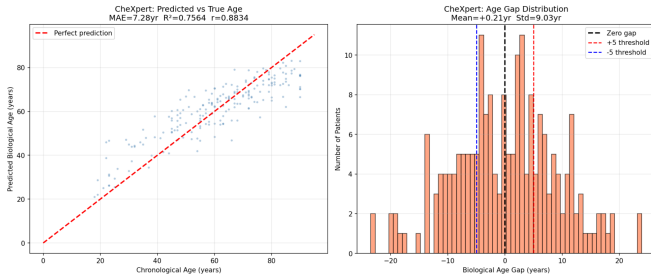


Fig. 6. Cross-dataset validation on CheXpert ($n = 202$). *Left*: Predicted versus chronological age scatter plot ($r = 0.8834$, $R^2 = 0.7564$), showing that the learned relationship remains meaningful under zero-shot transfer. *Right*: Biological age-gap distribution (mean bias = +0.21 years). The wider spread relative to the NIH test set likely reflects both the smaller sample size ($n = 202$) and the expected domain shift across institutions.

F. Comparison with Existing Methods

Table VIII provides a contextual comparison with related literature.

G. Sample Clinical Reports

The pipeline also generates structured patient-level summaries. For example, **Patient A** (chronological age 81 years, predicted biological age 73.5 years, age gap -7.45 years, Low risk) shows a biological age well below chronological age, which is consistent with comparatively favorable aging

TABLE VIII
CONTEXTUAL COMPARISON WITH EXISTING METHODS.

Method	Backbone	Reported metric	Modality
Raghu et al. [12]	CNN	Prognostic age-gap analysis (no MAE reported)	Chest X-Ray
Rajpurkar et al. [7]	DenseNet-121	Radiologist-level pneumonia detection benchmark (not age-regression)	Chest X-Ray
Khan et al. [14]	EfficientNet-B1	Accuracy = 96.13% in 4-class CXR classification (not age-regression)	Chest X-Ray
Park et al. [15]	ViT	Multi-hospital external validation for chest X-ray diagnosis (not age-regression)	Chest X-Ray
Iglovikov et al. [9]	U-Net/CNN	Bone-age error reported in months (not directly comparable)	Hand/Wrist X-Ray
Levine et al. [3]	Regression	Mortality-linked phenotypic aging score (not imaging MAE)	Blood biomarkers
BioAge-Net	ResNet-50	MAE = 4.91 years	Chest X-Ray

Note: A direct comparison of mean absolute error (MAE) across different methods is constrained due to variations in datasets, patient populations, and imaging modalities. Consequently, the comparison of underlying frameworks should be understood as contextual rather than as a precise quantitative evaluation.

status and supports continued preventive care. **Patient B**: The individual, with a chronological age of 74 years and a predicted biological age of 69.2 years, resulting in an age gap of -4.79 years and classified as Moderate risk, demonstrates a close concordance between biological and chronological age. Consequently, routine follow-up is deemed the appropriate course of action.

VI. CONCLUSION

Using a ResNet-50 model with a specially designed regression part and trained on the NIH ChestX-Ray14 dataset, the system shows strong performance when using just chest X-ray images. It achieved a mean absolute error of 4.91 years, root mean square error of 6.43 years, R^2 value of 0.8455, Pearson correlation of 0.9198, and a small mean bias of +0.24 years. These results show that chest X-ray images have enough

information about age to estimate biological age in a way that is useful for clinical purposes at a large scale.

Beyond age regression, the framework also uses a three-tier system to assess risk based on age gaps and includes Grad-CAM for explaining results, which improves how well the model's outputs fit into preventive screenings and electronic health records. A first test done on the CheXpert validation group ($n = 202$) showed an MAE of 7.28 years and a Pearson r of 0.8834, suggesting that the model works well across different medical centers even when there are changes in the data sources.

Even though, using this in real clinical settings needs a lot of testing in different groups of people, various imaging tools, and different ways of collecting images. Given that the training dataset comprises NIH Clinical Center radiographs collected between 1992 and 2015, the model should be regarded as a valuable screening adjunct rather than an independent diagnostic instrument. In clinical practice, the biological age gap metric is optimally interpreted in conjunction with other clinical indicators following site-specific calibration, multi-institutional evaluation, and regulatory approval.

VII. FUTURE SCOPE

Several future directions remain important:

- 1) **Multi-Modal Fusion:** Combining chest radiographs with variables such as blood pressure, body mass index, and laboratory findings may reduce MAE below three years.
- 2) **Pathology-Stratified Modeling:** Separate cardiac, pulmonary, and healthier-population models may help disentangle disease effects from aging signals.
- 3) **Longitudinal Validation:** Serial radiographs should be used to test whether age-gap changes correlate with mortality, hospitalization, and disease progression.
- 4) **Transformer Architectures:** Vision Transformers and Swin Transformers should be benchmarked as alternative backbones for radiographic age estimation.
- 5) **Mobile and Edge Deployment:** Distillation to lightweight models such as MobileNetV3 may support portable X-ray deployment in resource-constrained settings.
- 6) **Cross-Dataset Validation:** Larger external studies on MIMIC-CXR and PadChest are needed beyond the preliminary CheXpert result ($n = 202$; MAE = 7.28 years; $r = 0.8834$).
- 7) **Architectural Enhancements:** Applying squeeze-and-excitation mechanisms after global average pooling could further improve feature accuracy.
- 8) **Ablation Studies:** Thorough ablation studies with confidence intervals and paired t -tests are needed to measure the contribution of each design component.

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